

JavaScript Lab 1

Objective

This lab is designed to help you deeply understand JavaScript fundamentals, variable declarations, scope, type coercion, and control flow structures.

Note: Try all code blocks and discuss the code with your lab instructor.

Part 1: HTML Setup

Task 1.1:

Create an HTML file ('index.html') containing at least the following structure:

```
...
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>JavaScript Lab 1</title>
</head>
<body>
  <h1>JavaScript Lab 1</h1>
  <!-- Scripts will be added here -->
</body>
</html>
...
```

Part 2: External JavaScript File

Task 2.1:

Create an external JavaScript file named 'index.js' and define the following:

```
...
// a. Number variables
var number1 = 3;
var number2 = 2.9;
var number3 = 0xff; // Hexadecimal 255
```

```
// String variables with different quotes
var firstName = "John";
var middleName = 'Doe';
var lastName = `Smith`;

// b. Boolean flag
var flag = true;

// c. Log message
console.log("This is the External JavaScript file");
...

---
```

Part 3: Variable Scope & Behavior Experiments

Experiment Set A: Hoisting & Declaration

1. In `External.js`, try printing `number1` **before** its declaration.
2. In **HTML**, add an internal `<script>` block **before** linking `External.js` and print `number1`.
3. In **HTML**, add another internal `<script>` block **after** linking `External.js` and print `number1`.

Repeat steps 1–3 for `number2` and `number3`.

Experiment Set B: `var` vs. Implicit Globals

4. Remove the `var` keyword from `number1` and repeat Experiment Set A.
Observe and note the differences.

Experiment Set C: Mutability & Type

5. In the browser console, try to reassign:

```
...
number1 = 100;
...
```

6. Try to modify a string character:

```
...
firstName[3] = 'A';
console.log(firstName);
...
```

7. Print the `typeof` for all variables.

Part 4: Type Coercion & Operators

Inside an **embedded `<script>` block** in `index.html`, perform and log the following:

```

|-----|-----|-----|
| a. Numeric Addition | `number1 + number2` ||
| b. Boolean + Number | `flag + number1` ||
| c. String + Boolean | `firstName + flag` ||
| d. Number + String | `number1 + firstName` ||
| e. Mixed Addition | `number1 + number2 + firstName` ||
| f. String Concatenation Order | `number1 + firstName + number2` ||
| g. Number × Boolean | `number1 flag` ||
| h. Number ÷ String | `number1 / lastName` ||

```

Template Literals (ES6)

i. Print your full name using backticks:

```

...
console.log(`Full Name: ${firstName} ${middleName} ${lastName}`);
...

```

Console Table

j. Display an array as a table:

```

...
const data = [
  { id: 1, name: "Alice" },
  { id: 2, name: "Bob" }
];
console.table(data);
...

```

Part 5: Control Flow & Loops

Task 5.1: Odd/Even Check

Write a script to check if a given number is odd or even.

Task 5.2: Loop (1–10)

Print numbers 1 to 10 using loops.

Task 5.3: Sign Check

Check if a number is negative, positive, or zero.

Task 5.4: Multiplication Table

Print multiplication table for any number in the console.

Task 5.5: Day of Week

Write a program that takes a number from 1 to 7 and prints the corresponding day of the week (e.g., 1 -> "Sunday", 2 -> "Monday", etc.).

Task 5.6: Weekend/Weekday

Write a program that takes a day number and prints whether it's a weekend or a weekday.

Part 6: Search Questions

Q1: Strict Mode

Research: In ES5, we can avoid defining variables without `var` keyword by using 'use strict' mode. How does this work?

Q2: `toString(radix)` Method

All number variables have a method called `toString`. For example:

...

```
var number = 30;  
number.toString(16);  
...
```

What does this method do?
